

Transformer vs. Physics-Inspired Machine Learning for Particle Data

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- At the Large Hadron Collider, high-energy pp and heavy-ion collisions produce particle jets measured by large detectors
- We use provided JEWEL Monte Carlo jet samples: vacuum-like reference jets and reconstructed medium-affected jets
- We test whether ML models can identify medium-induced jet modifications from particle-level constituents
- We compare physics-inspired Energy Flow Network architectures with self-attention and state-space based models, and see how these compare in performance, stability, and interpretability

What is a QCD Jet?

- QCD = Quantum Chromodynamics
- Quarks and gluons are confined by the strong nuclear force
- They form color-neutral hadrons
- A QCD jet is a collimated spray of hadrons
- QCD Jets come from energetic quarks or gluons
- QCD Jets are measured in particle detectors

[1, 2, 9]

Physics Background II

Particle features used in the project: $\mathbf{x}_i = (z_i, \Delta\eta_i, \Delta\phi_i)$

– z_i : momentum fraction of one particle

➤ $z_i = \frac{P_{T,i}}{P_T^{jet}}$, P_T : transverse momentum

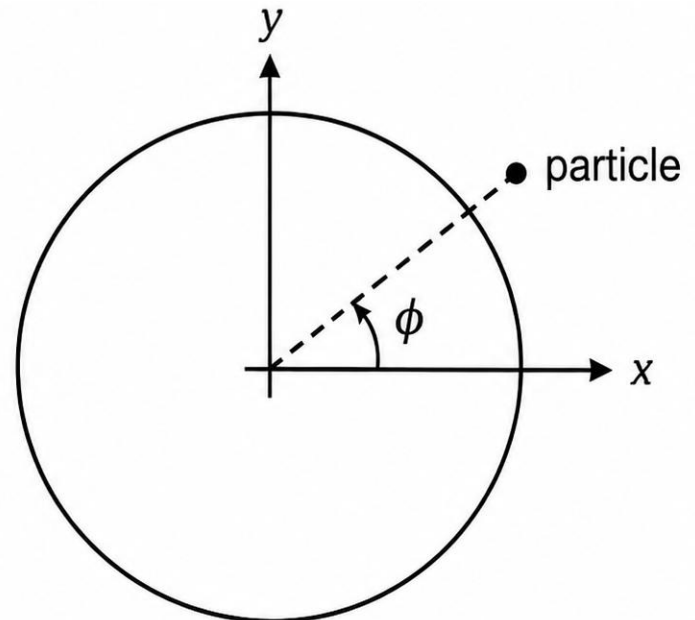
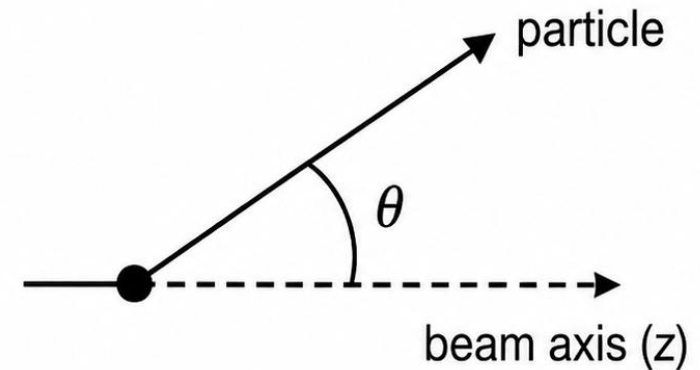
– η_i : pseudorapidity, direction along the beam axis

➤ $\eta = -\ln\left(\tan\frac{\theta}{2}\right)$

– Polar Angle (θ): The angle relative to the beam axis

– ϕ_i : azimuthal angle around the beam axis

➤ $\Delta\phi_i = \phi_i - \phi_{jet}$

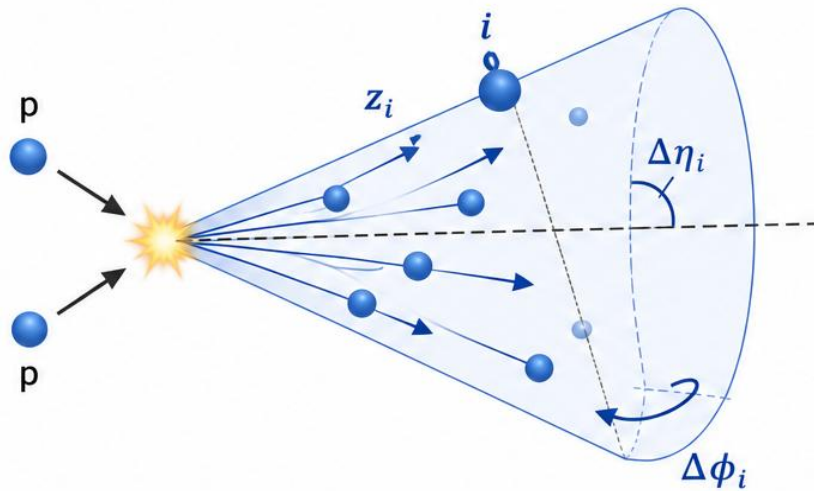


[9]

pp vs. PbPb Collisions

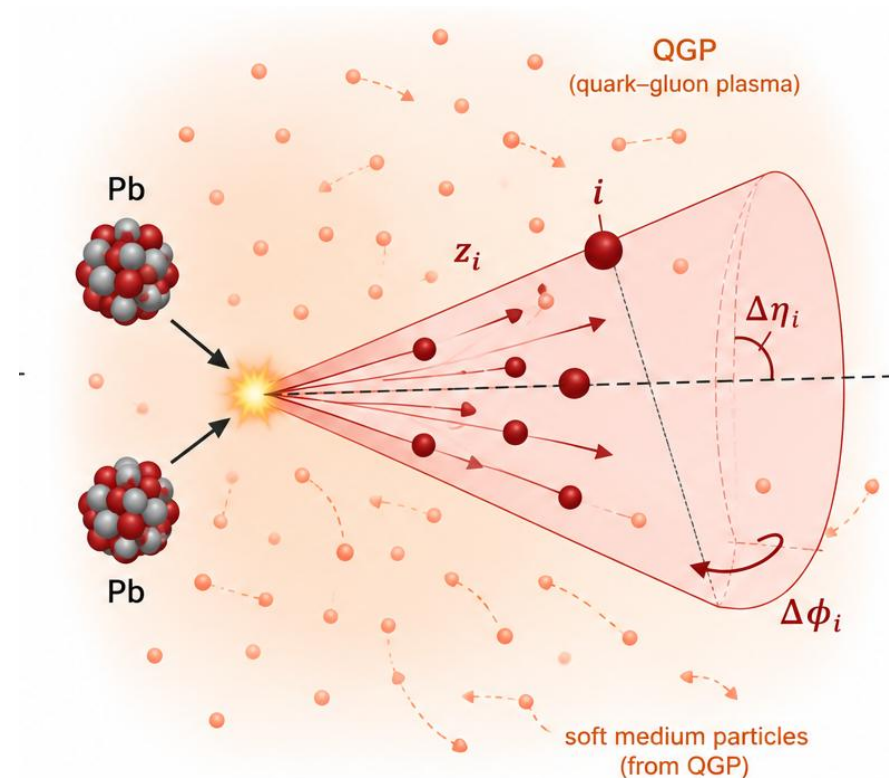
pp collisions:

- Proton-proton collisions
- Used as a reference sample
- Jets are closer to vacuum-like jets in heavy-ion collisions

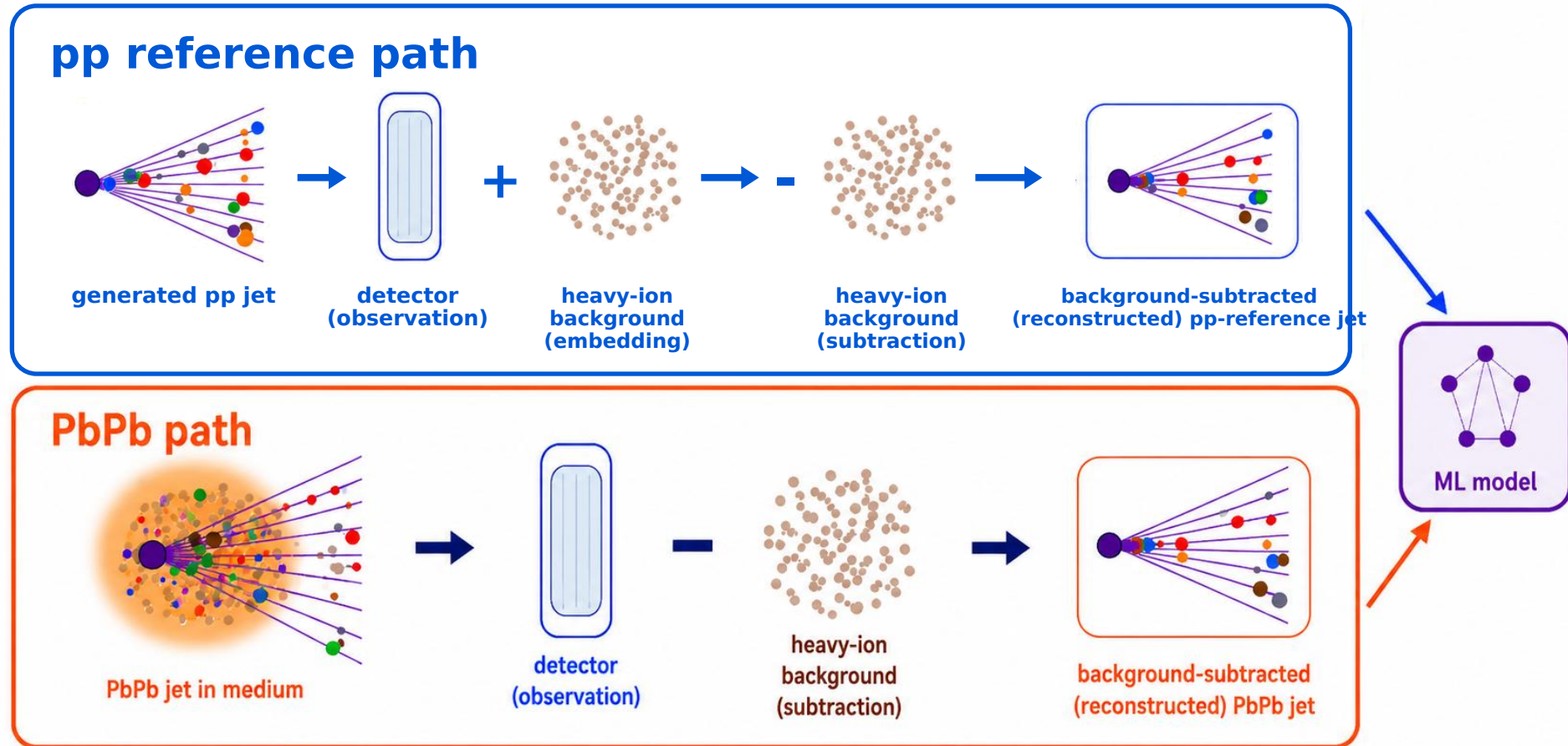


PbPb collisions:

- Lead-lead heavy-ion collisions
- Create a hot and dense medium
- This medium is called the quark-gluon plasma



Apples-to-Apples Setup



- Both jet samples are reconstructed in the same way, so the ML model can compare them fairly

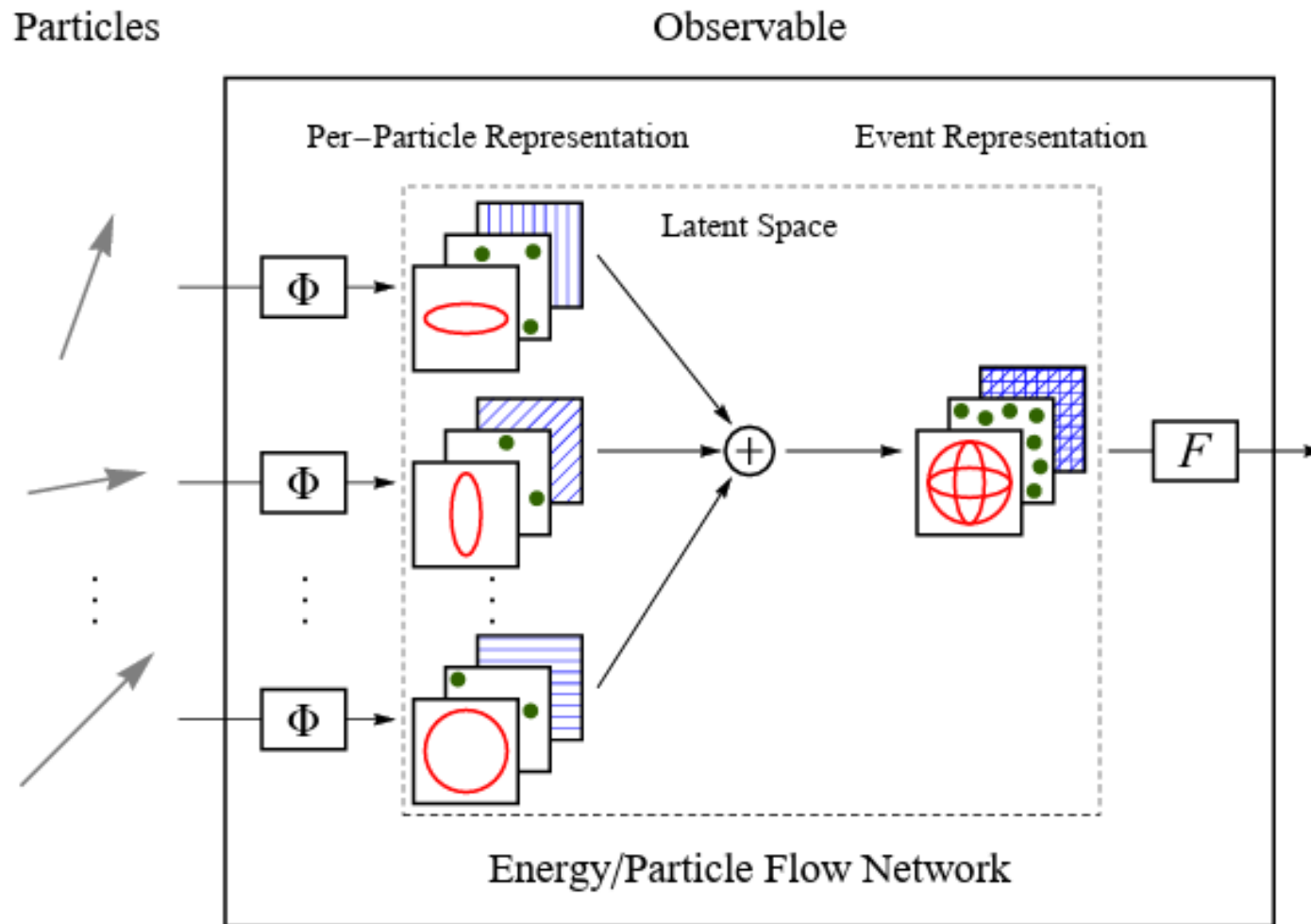
[1]

Research Question

Which model family performs better for pp vs. PbPb jet classification: attention- or state-space-based architectures or physics-inspired EFN-style models?

Models

Energy Flow Network



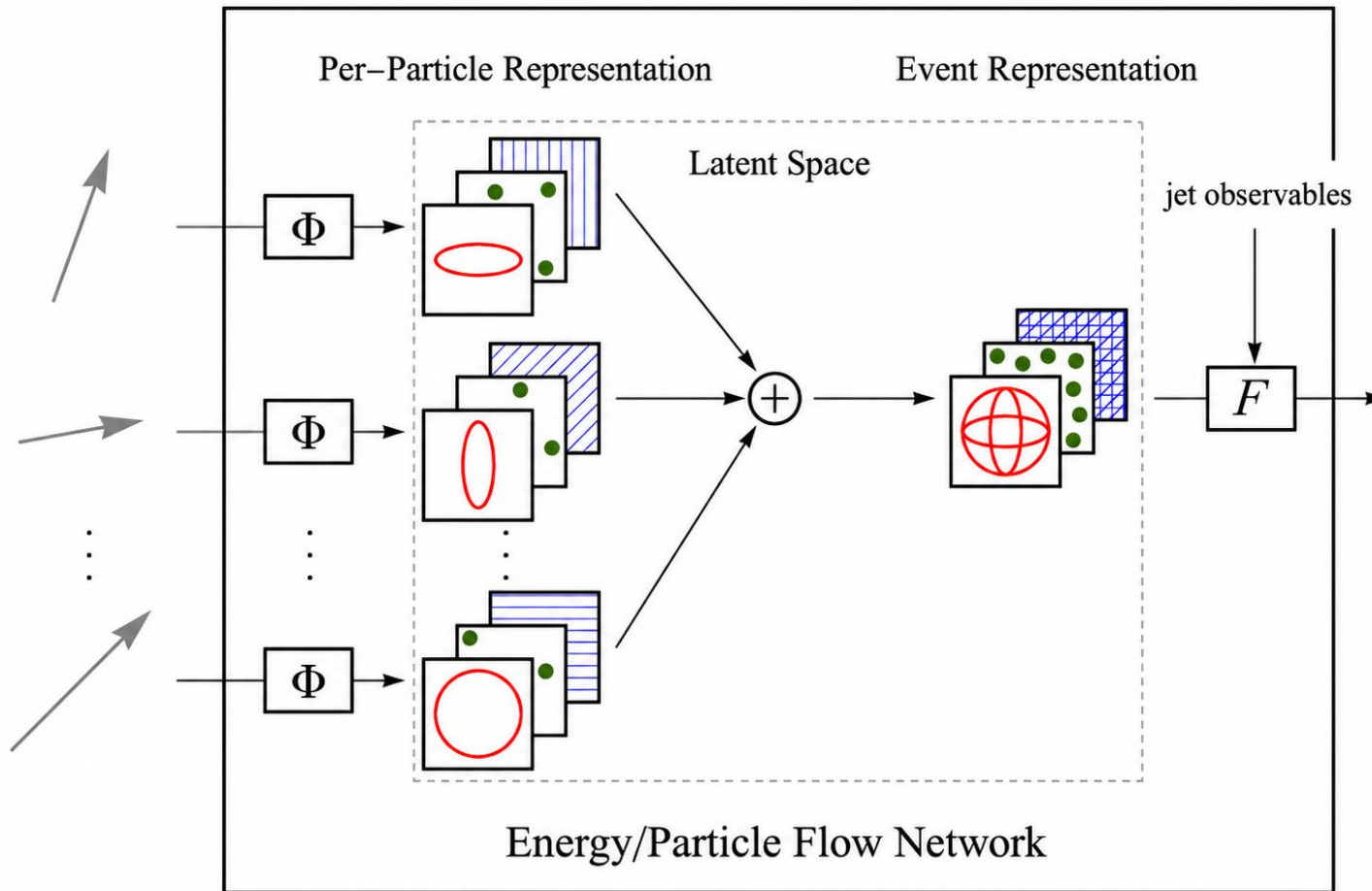
- Input: particles in a jet
- Φ learns a representation for every particle
- Sum creates one jet representation
- F predicts the class
- Formula:
 - $O(P) = F(\sum_i z_i \phi(\hat{p}_i))$
 - $O(P)$ = model output for the particle set P

- First- and higher-order moment representation
 - Formula:
 - $O_k(P) = F \left(\langle \phi^\alpha \rangle_P, \langle \phi^\alpha \phi^b \rangle_P, \dots, \langle \phi^\alpha \dots \phi^k \rangle_P \right)$
 - P = set of particles in one jet
 - $O_k(P)$ = model output for the jet
 - K = maximum moment order
- Higher moments capture multi-particle correlations and more detailed jet structure

Observable Energy Flow Networks

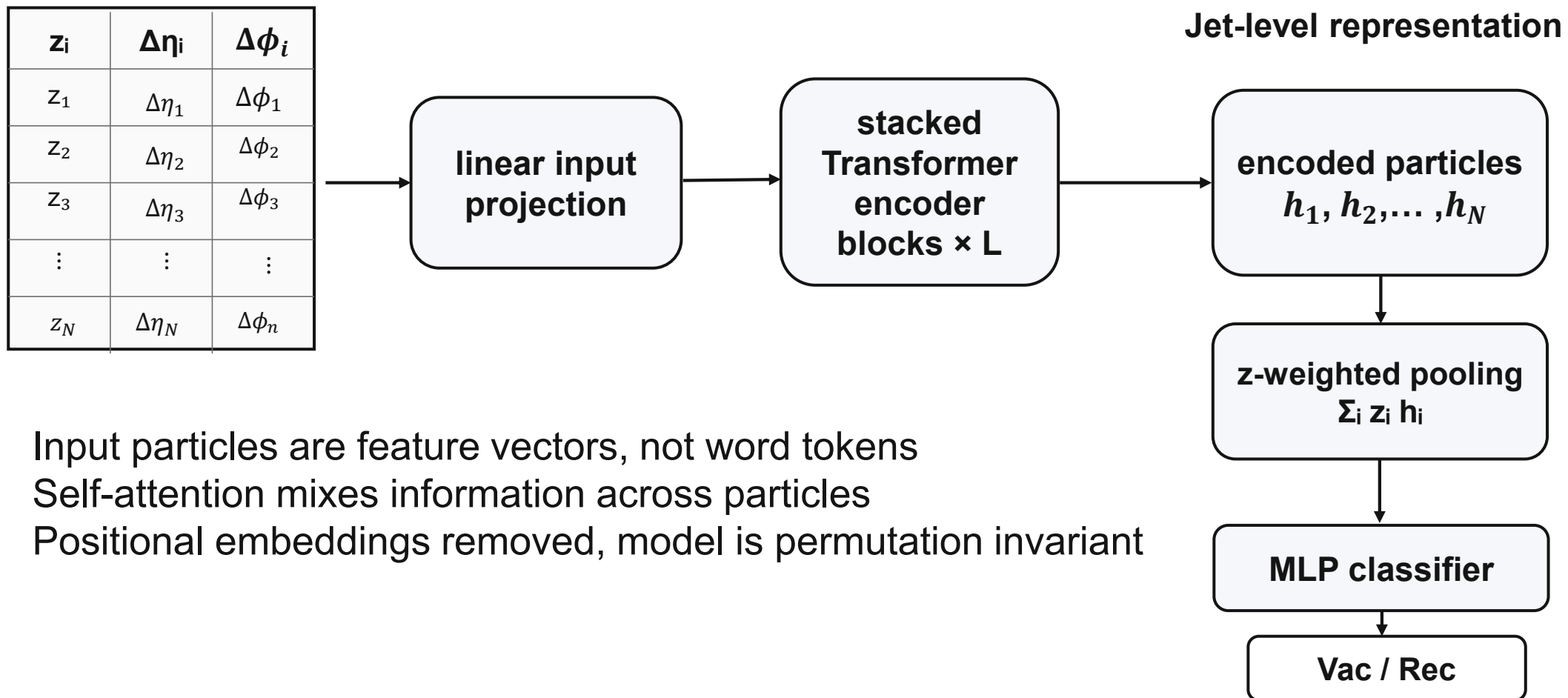
Particles

Observable



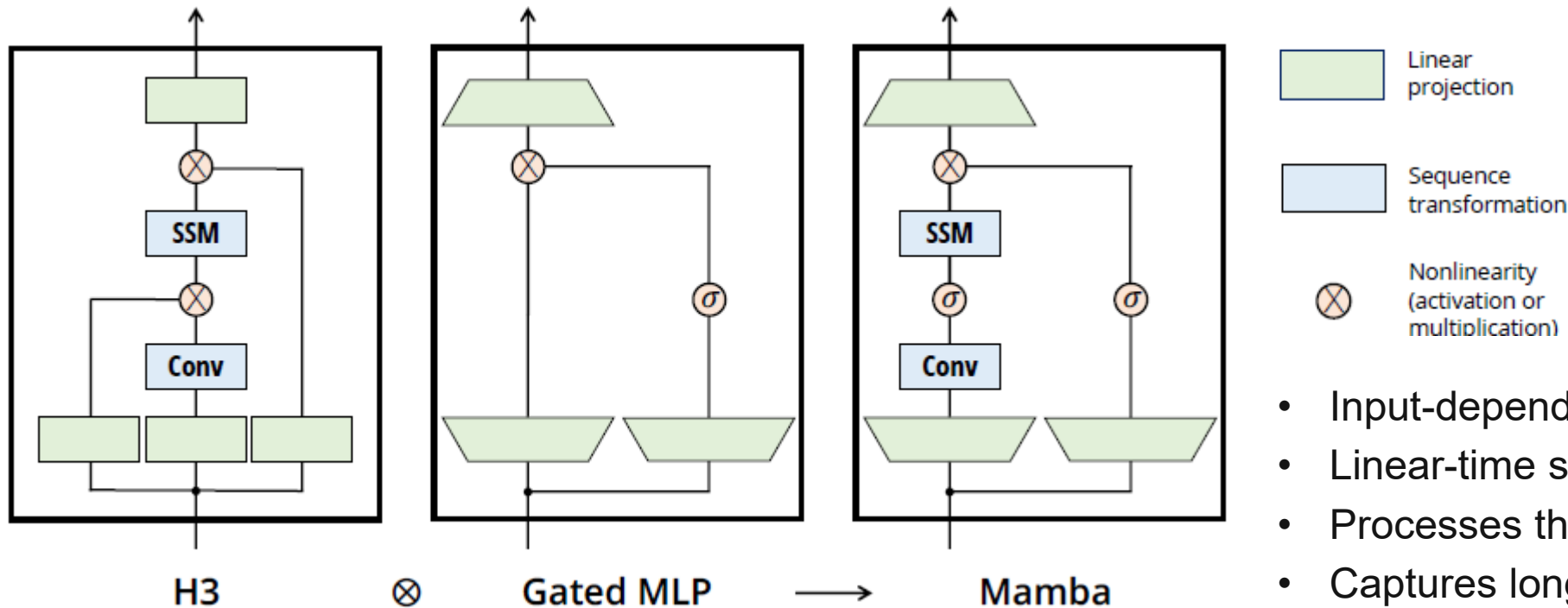
- Goal: combine learned EFN features with known physics features
- Formula:
- $O(P) = F \left(\left[\sum_i z_i \phi(\hat{p}_i), o \right] \right)$
- o = N-subjettiness observables + Energy Flow Polynomials

Transformer Encoder



- Input particles are feature vectors, not word tokens
- Self-attention mixes information across particles
- Positional embeddings removed, model is permutation invariant

Mamba / Selective State-Space Model

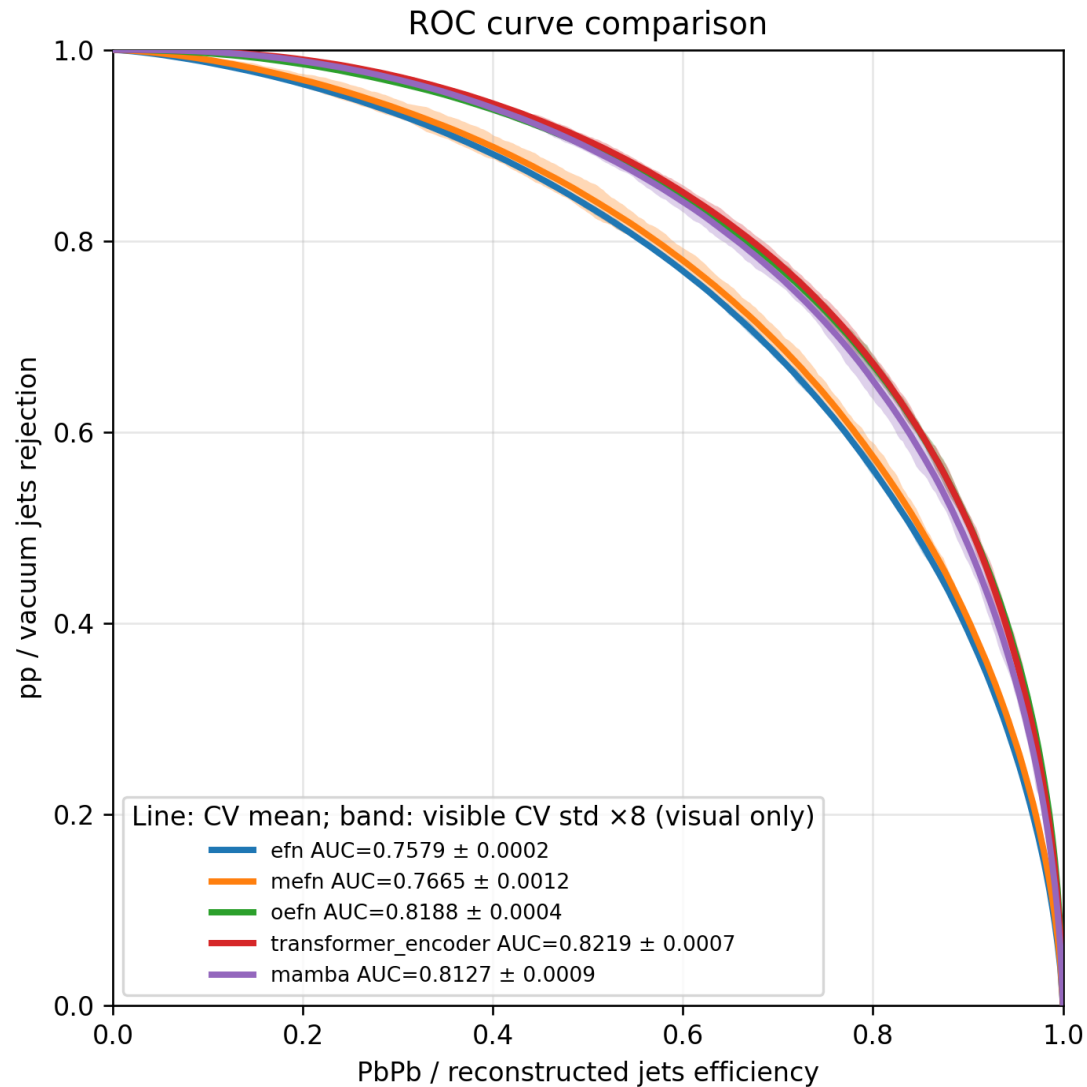


- Input-dependent gating
- Linear-time sequence processing
- Processes the jet as a particle sequence
- Captures long-range particle dependencies
- More memory efficient than self-attention

Input: ordered particle features $x_i = (z_i, \Delta\eta_i, \Delta\phi_i) \rightarrow$ Mamba $\rightarrow$ jet representation $\rightarrow$ classifier

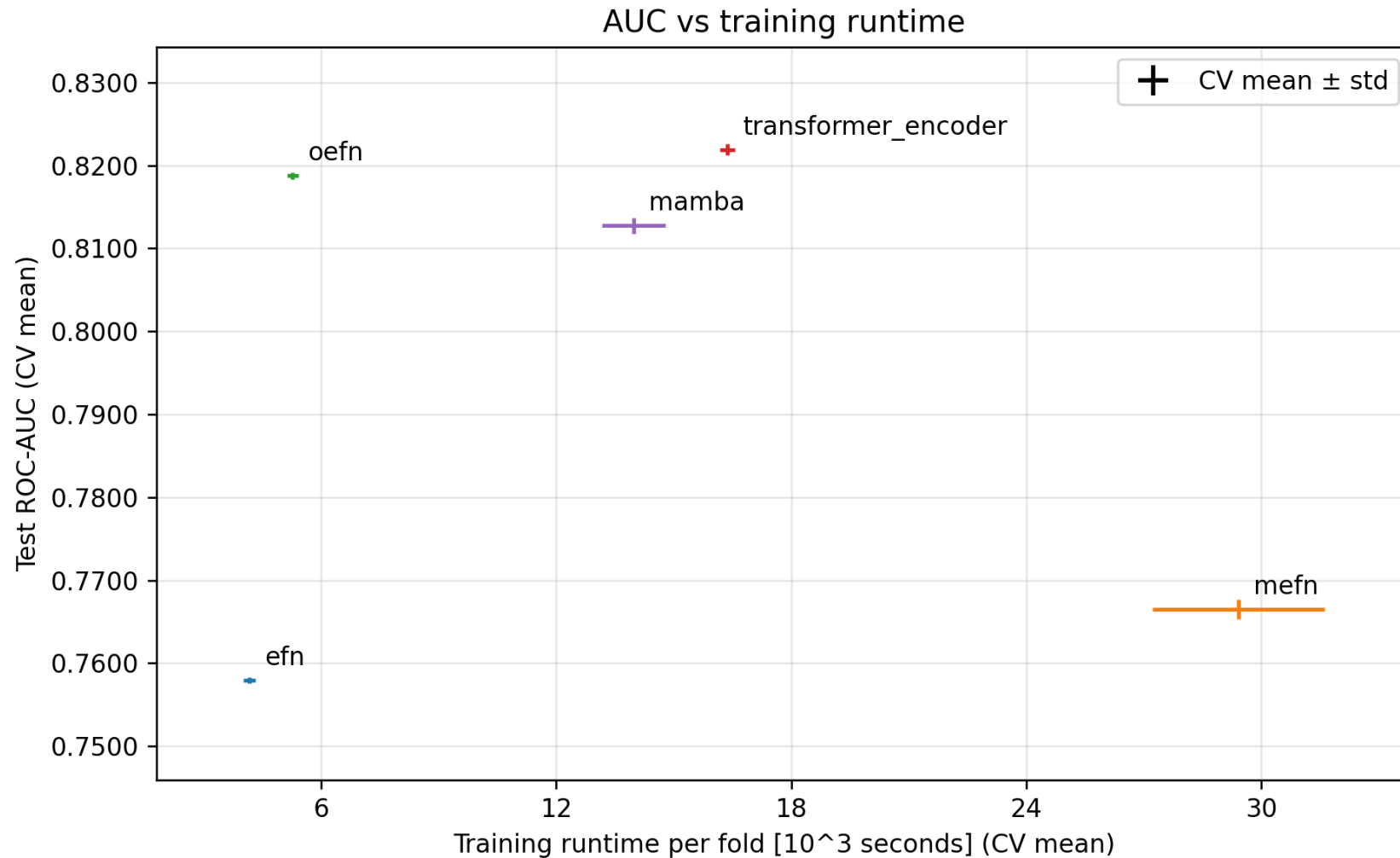
Experiment

Experiment – Results I



- The ROC curves show that Transformer and oefn perform best, while efn and mefn show lower AUC values in this comparison

Experiment - Results II



➤ The Transformer reaches the highest AUC, while oEFN is close with much lower training time

- Transformer achieves the highest classification performance
- oEFN achieves essentially the same performance as the Transformer, but with much lower training cost
- Physics-informed EFN extensions remain competitive with general neural architectures
- No single architecture is best across both predictive performance and efficiency

- Attention-EFN, observable-aware attention, and Observable-Mamba variants
- Optimize model architectures and jet-level pooling strategies
- Run additional hyperparameter tuning, robustness checks across folds and particle orderings

- [1] J. A. Gonçalves and J. G. Milhano. “Apples to apples in jet quenching.” JHEP 09 (2025) 028
- [2] P. T. Komiske, E. M. Metodiev, J. Thaler. “Energy Flow Networks: Deep Sets for Particle Jets.” JHEP 01 (2019)
- [3] R. Gambhir, A. Osathapan, J. Thaler. “Moments of Clarity: Streamlining Latent Spaces in Machine Learning using Moment Pooling.” arXiv:2403.08854.
- [4] J. Thaler and K. Van Tilburg. “Identifying Boosted Objects with N-subjettiness.” JHEP 03 (2011) 015
- [5] P. T. Komiske, E. M. Metodiev, J. Thaler. “Energy Flow Polynomials.” JHEP 04 (2018) 013
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- [7] J. Devlin et al. “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding.” arXiv:1810.04805.
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- [9] J. A. Gonçalves. “All is More: Energy Flow Networks for Jet Quenching.” arXiv:2510.26879.



We are happy
to answer any
questions

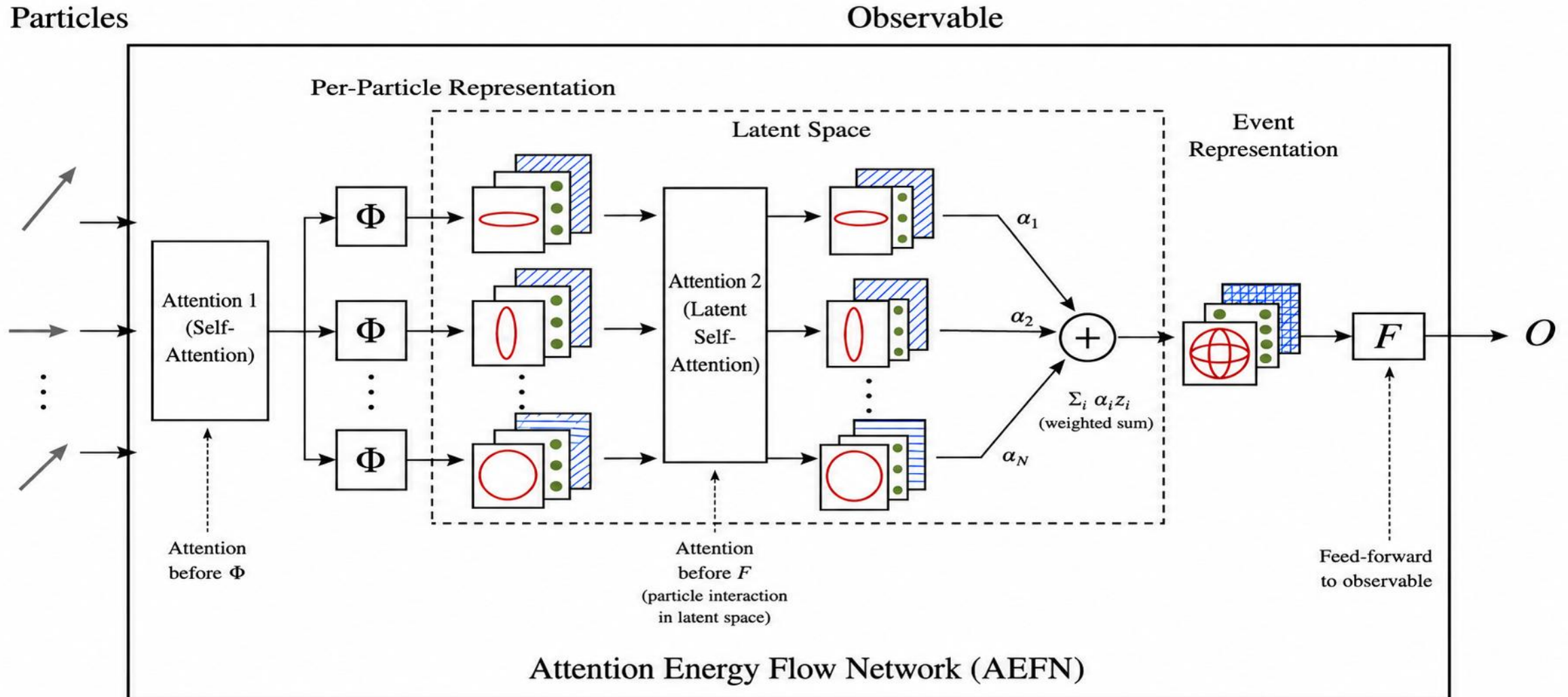


Backups

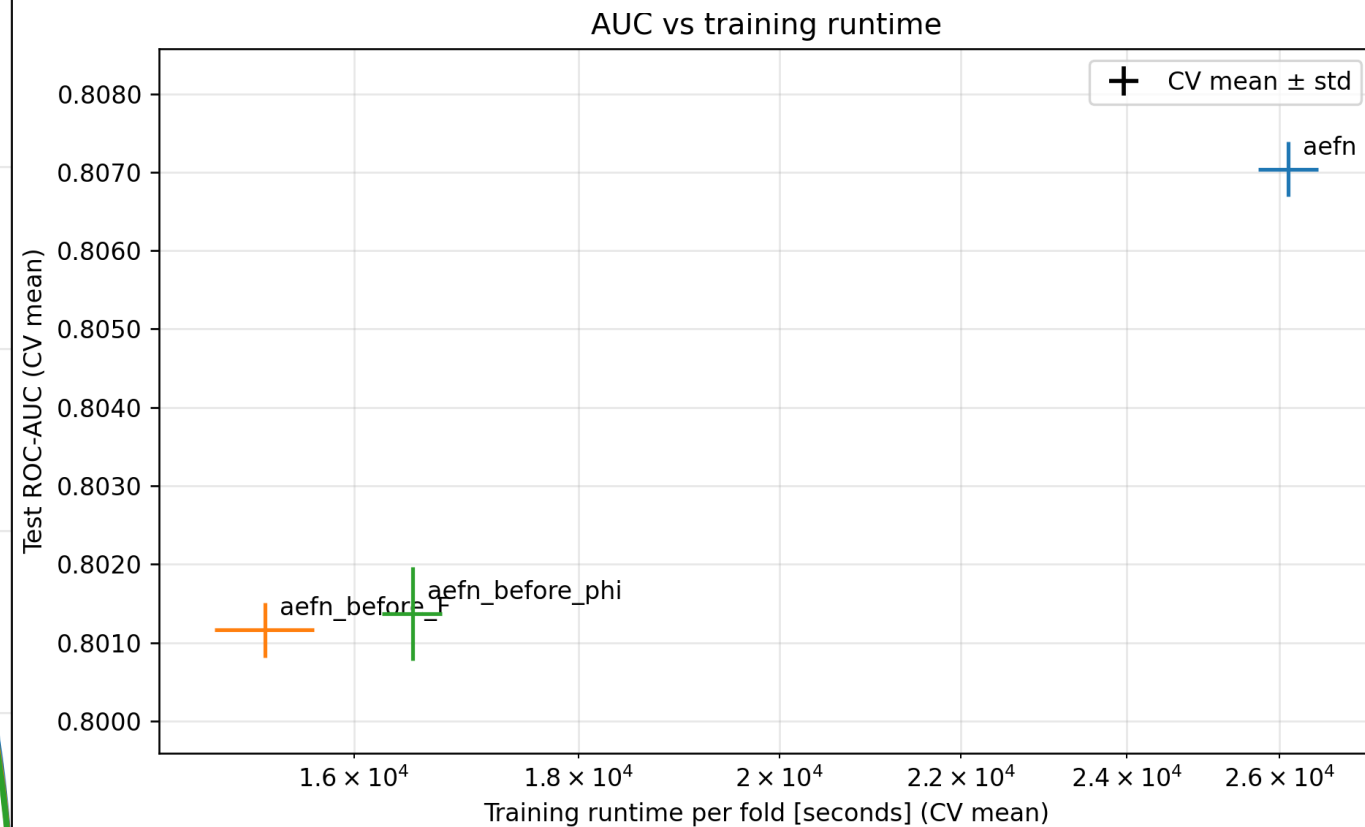
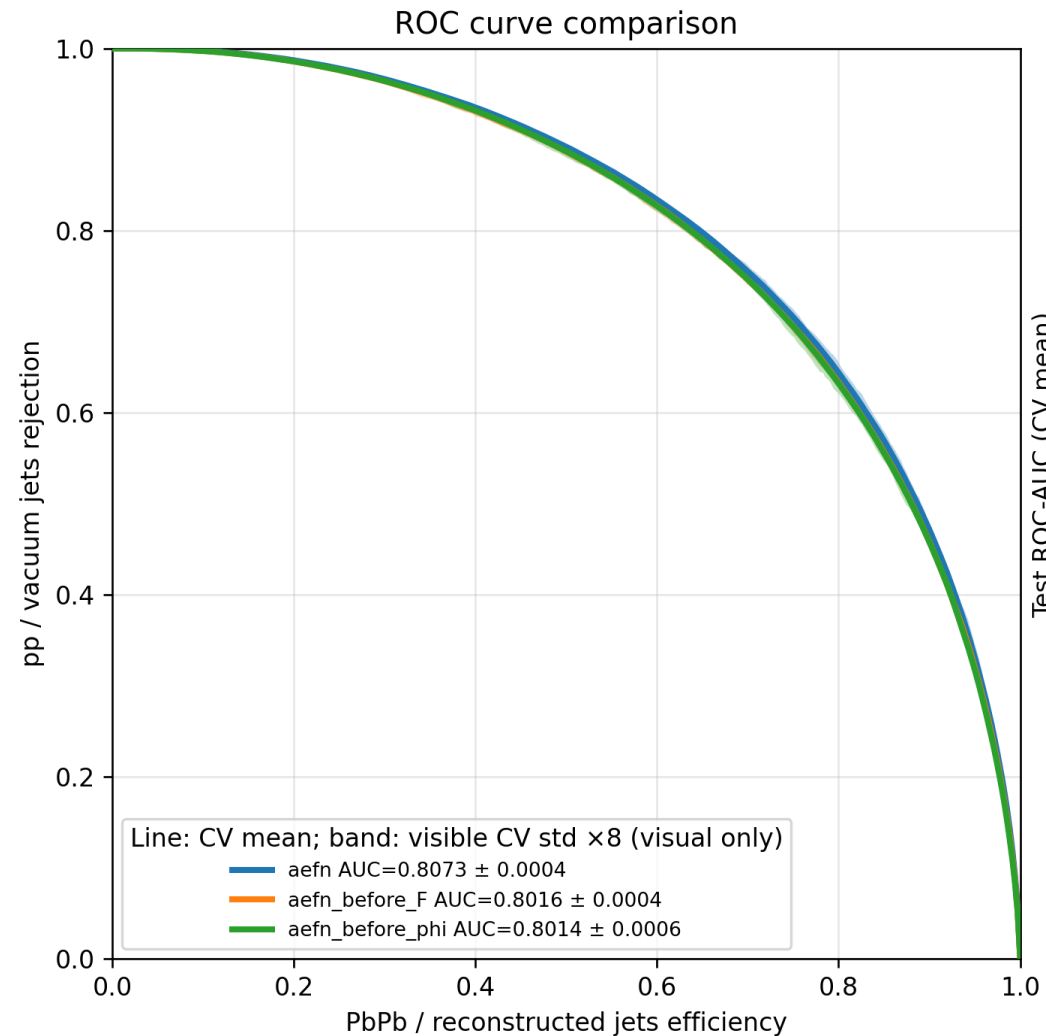
Shared run configuration

Parameter	Value
num_data	-1 (all available data)
max_particles	128
Max_files_per_class	-1
batch_size	512
epochs	200
patience	30
learning_rate	0.0001
weight_decay	0.0001
num_folds	4
final_test_ratio	0.2
seed	42
dropout	0.1
activation	Silu

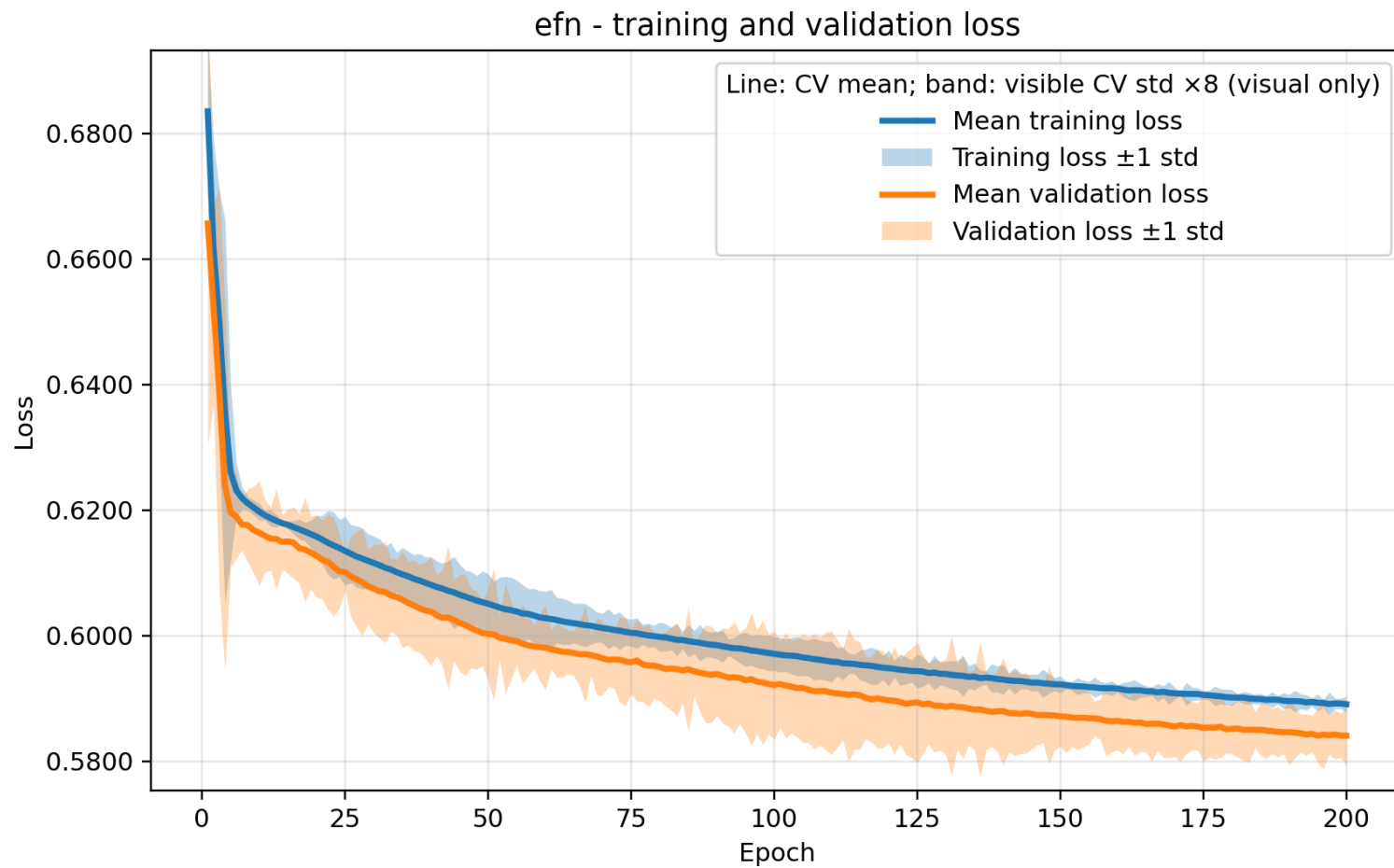
Attention Energy Flow Network



AEFN - Results

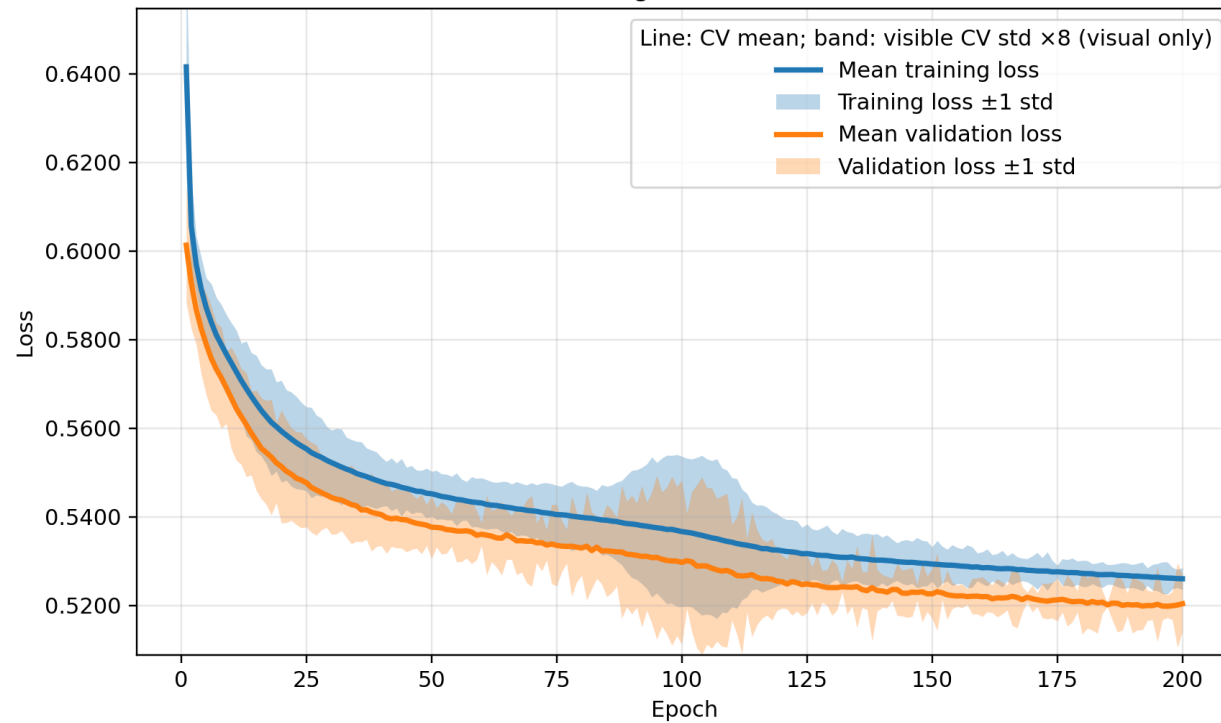


Baseline – Results I

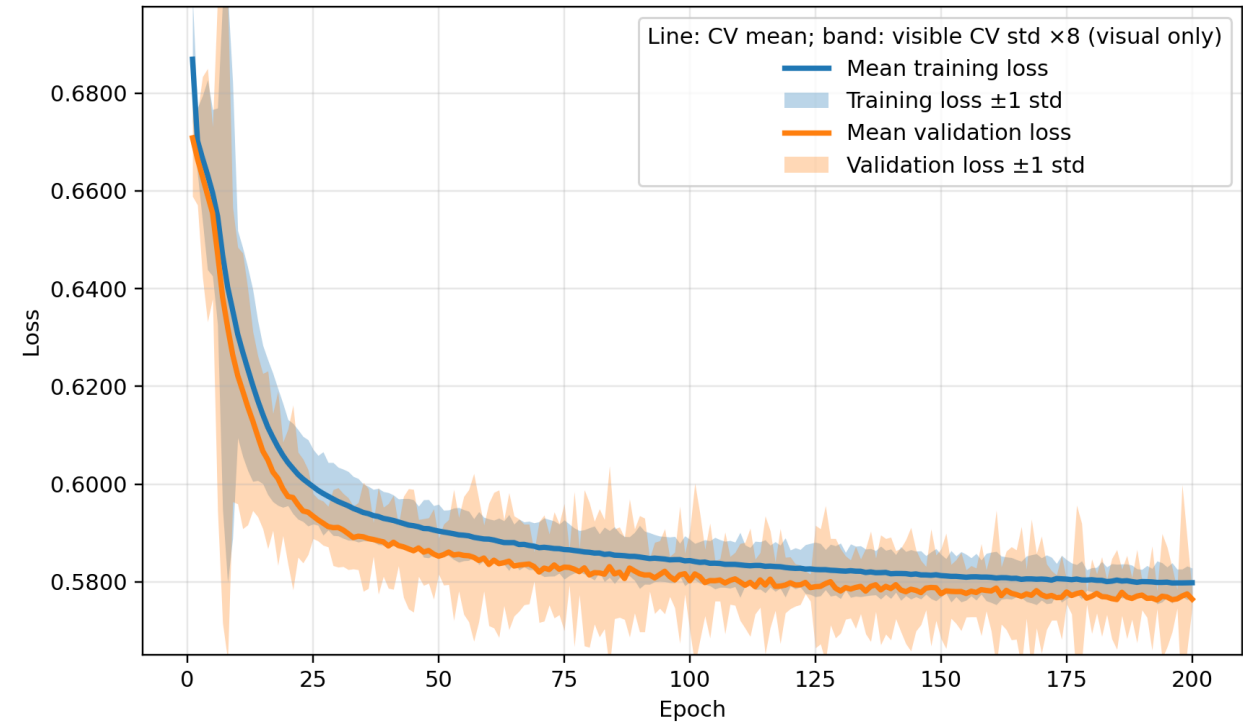


Baseline – Results II

oefn - training and validation loss

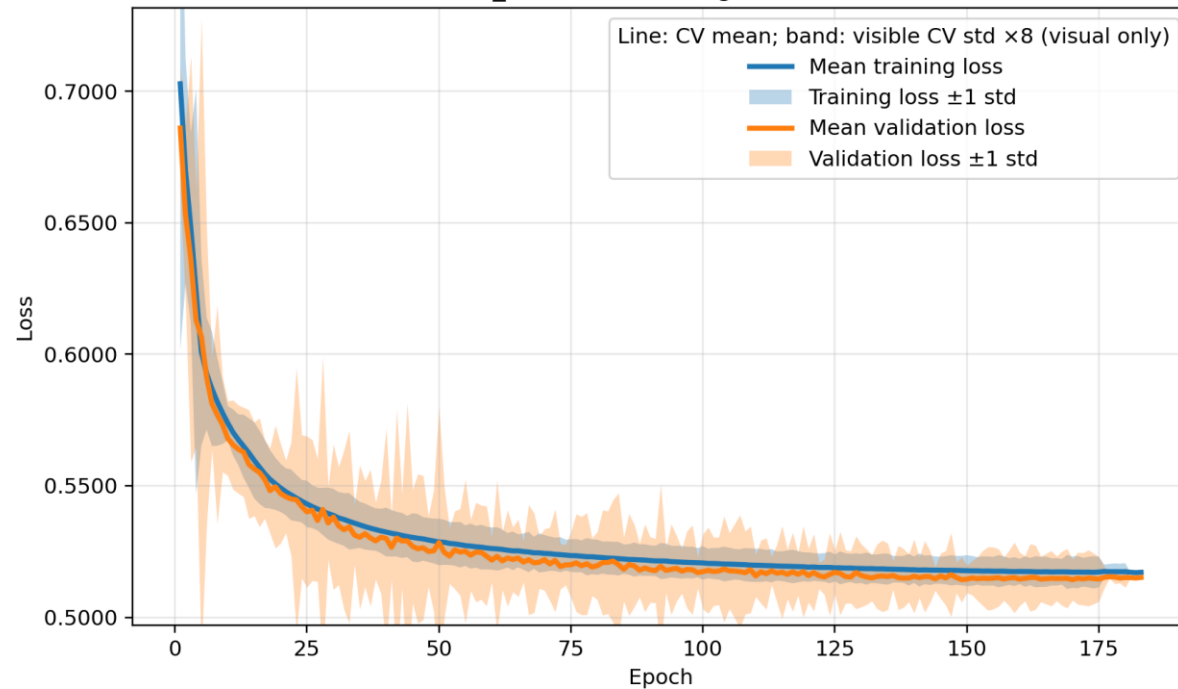


mefn - training and validation loss



Baseline – Results III

transformer_encoder - training and validation loss



mamba - training and validation loss

